

CS 4910 Final Submission

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Course: CS 4910 Senior Design

Project: IGVC Object Detection / Perception

Overview

This repository contains my final submission materials for CS 4910. My work focused on the object detection/perception portion of the IGVC senior design project.

The perception system uses camera-based object detection to identify important objects for autonomous navigation, including obstacles such as barrels, potholes, and signs.

Submission Contents

This repository includes the required final submission materials:

1. Test Report
2. Requirements Document
3. Design Document
4. Final Implementation
5. Traceability Information

Repository Structure

```
CS4910_Final_Submission/  
|  
|-- README.md  
|-- Requirements.md  
|-- Design.md  
|-- Traceability.md  
|-- Test_Report.md  
|-- Final_Implementation.md  
|-- requirements.txt  
|  
|-- src/  
|   |-- detect_image.py  
|   |-- detect_video.py  
|   |-- export_model.py  
|   |-- benchmark_models.py  
|  
|-- tests/  
|   |-- metamorphic_tests/  
|   |-- mutation_tests/
```

```
|  
`-- results/  
    |-- test outputs, screenshots, and supporting results
```

Running the Test Scripts

Install dependencies:

```
pip install -r requirements.txt
```

Run metamorphic tests:

```
python tests/metamorphic_tests/test_brightness.py  
python tests/metamorphic_tests/test_resize.py
```

Run mutation tests:

```
python tests/mutation_tests/test_confidence_thresholds.py  
python tests/mutation_tests/test_image_transformations.py
```

The scripts save output images and detection results in the results/ folder.